

Plant Disease Prediction

A Smart Approach to Modern Agriculture

Presented by: Neha Kumari

Introduction to Plant Disease Prediction



Agriculture

Foundation for global economies and livelihoods.



Impact of Diseases

Crop losses deeply affect farmers' income.



Early Detection

Timely intervention prevents widespread damage.



AI-Based Detection

Accurate analysis through leaf image processing.

Market Survey Highlights

Market Size

Global agri-tech to exceed $22B by 2025

Crop Loss

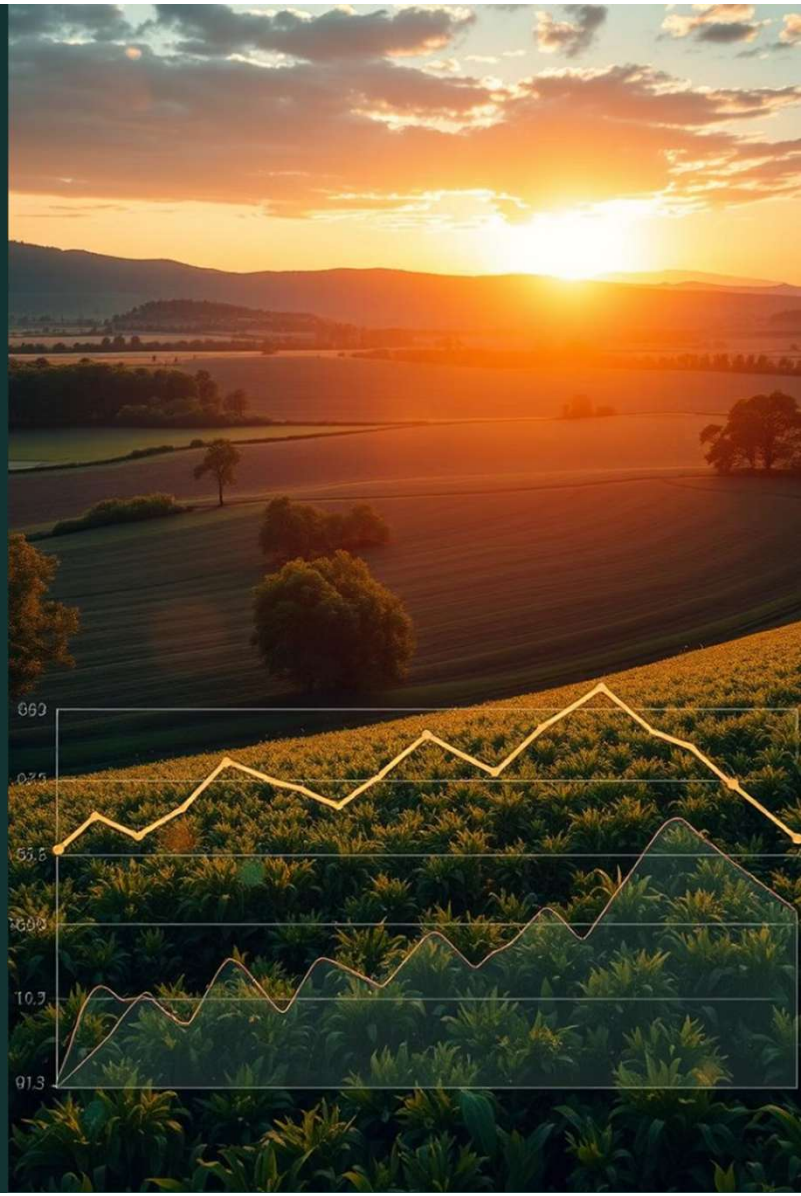
30–40% loss globally due to diseases

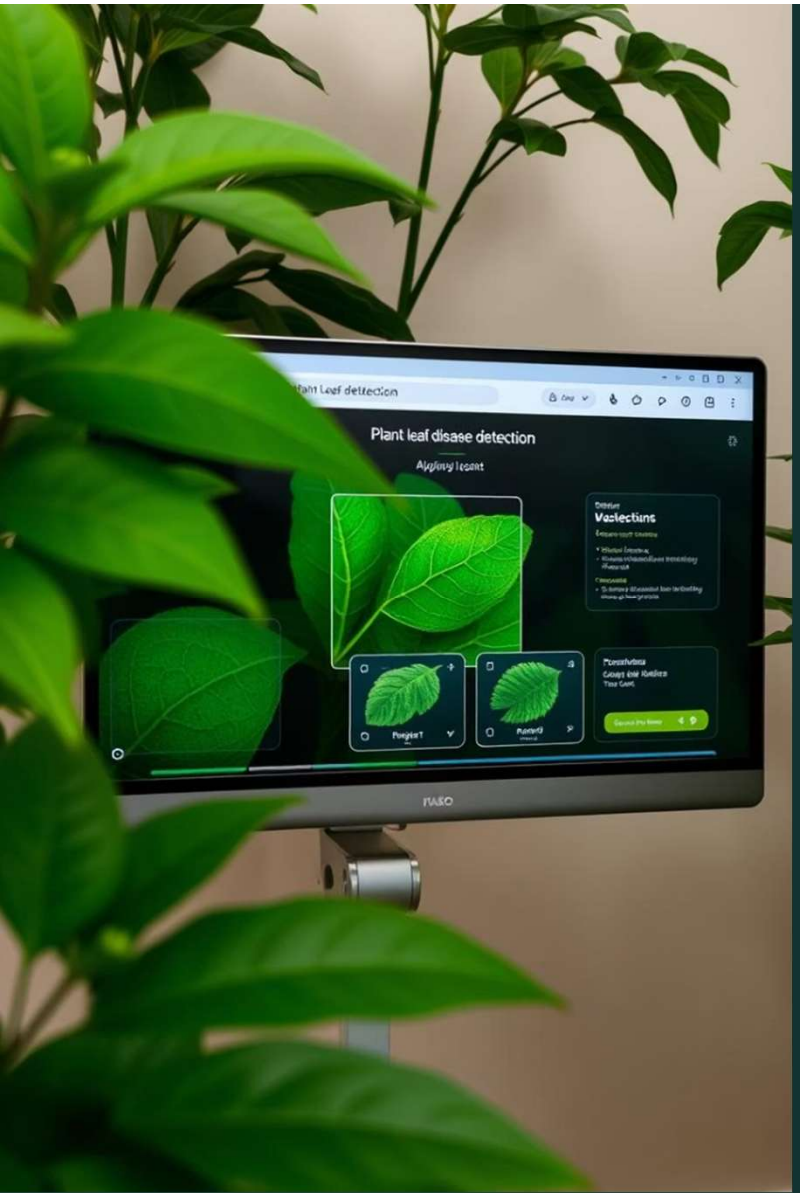
Demand

Growing need for smart farming tools

Farmer Needs

Fast, affordable, accurate detection





Proposed Solution Overview

CNN Model

Image-based disease classification

FastAPI Backend

Prediction and advice API (using Gemini)

React Frontend

Interactive and responsive UI

Weather Integration

Holistic plant health support

Project Phases

1

Requirement Analysis

User needs & disease types

2

Data Collection

Leaf image datasets from PlantVillage

3

Model Building

Custom CNN training

4

Backend Development

API with FastAPI

5

Frontend Development

React UI, Image Upload

6

Testing & Deployment(If required)

Dockerized, cloud ready

Model Architecture & Backend

CNN Model Details

- Input: RGB leaf images (resized)
- Conv + ReLU + MaxPooling layers
- Dense & Dropout for regularization
- Output: Softmax disease prediction
- Accuracy: ~95%

FastAPI Backend

- Processes images, normalization
- Runs CNN for predictions
- Uses GPT for mitigation advice
- Returns prediction & confidence JSON

Frontend Features

React + Bootstrap

Elegant and responsive design

Image Upload

Preview and process leaf images

Loading Spinner

Animated predictor feedback

Weather Widget

Live climate data integration

GPT Chat Advice

Natural-language disease guidance



Additional Features

Weather-based Forecast

Proactive Plant Health Insights

Responsive UI

Supports Dark Mode for Usability

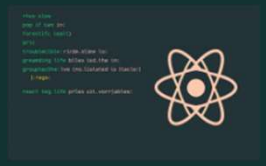
Environmental Data

Rainfall, humidity, wind speed displayed

5-day Forecast

Short-term climate outlook

Technology Stack



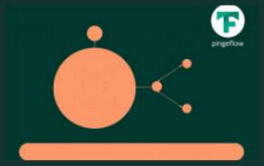
Frontend: React & Bootstrap

Interactive user interface built with React and styled with Bootstrap.



Backend: FastAPI with Python

High-performance API built with FastAPI for handling predictions.



Model: TensorFlow/Keras CNN

Convolutional Neural Network trained with TensorFlow/Keras for image analysis.



AI Integration: Gemini/GPT API

Natural language processing via Gemini/GPT API for intelligent advice.



Weather: OpenWeatherMap API

Real-time weather data integration using OpenWeatherMap API.



Future Scope

Expand Crop/Disease Support

Broader plant coverage

SMS Integration

Direct farmer notifications

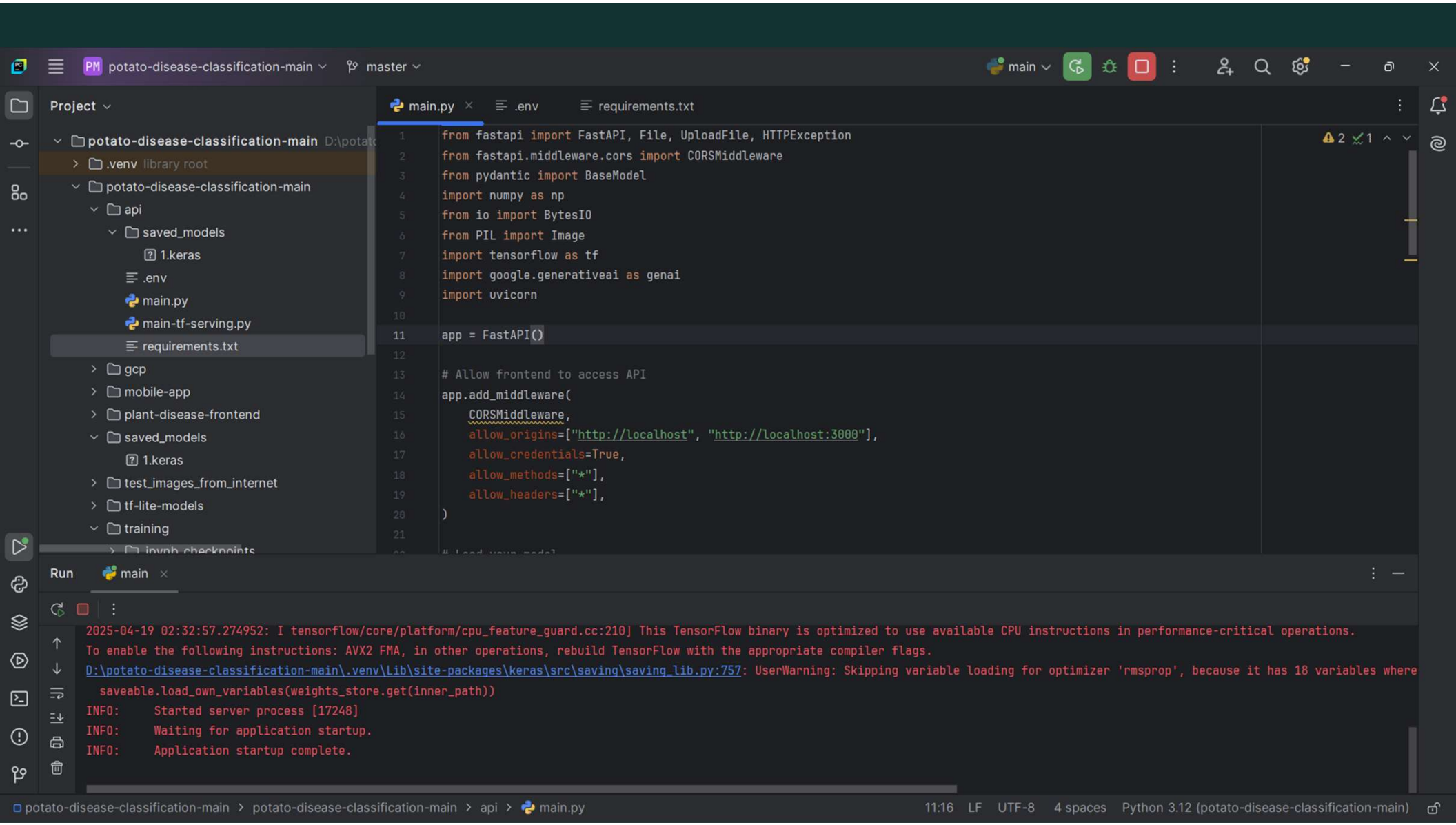
Mobile Real-time App

Camera capture and instant prediction

Satellite & IoT Sensors

Enhanced data for precision monitoring

BACK-END


potato-disease-classification-main
master
main
Project
potato-disease-classification-main D:\potato
.venv library root
potato-disease-classification-main
api
saved_models
1.keras
.env
main.py
main-tf-serving.py
requirements.txt
gcp
mobile-app
plant-disease-frontend
saved_models
1.keras
test_images_from_internet
tf-lite-models
training
main.py
.env
requirements.txt
from fastapi import FastAPI, File, UploadFile, HTTPException
from fastapi.middleware.cors import CORSMiddleware
from pydantic import BaseModel
import numpy as np
from io import BytesIO
from PIL import Image
import tensorflow as tf
import google.generativeai as genai
import uvicorn
app = FastAPI()
Allow frontend to access API
app.add_middleware(
CORSMiddleware,
allow_origins=["http://localhost", "http://localhost:3000"],
allow_credentials=True,
allow_methods=["*"],
allow_headers=["*"],
)
Run
main
2025-04-19 02:32:57.274952: I tensorflow/core/platform/cpu_feature_guard.cc:210] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
D:\potato-disease-classification-main\.venv\Lib\site-packages\keras\src\saving\saving_lib.py:757: UserWarning: Skipping variable loading for optimizer 'rmsprop', because it has 18 variables where
saveable.load_own_variables(weights_store.get(inner_path))
INFO: Started server process [17248]
INFO: Waiting for application startup.
INFO: Application startup complete.
potato-disease-classification-main > potato-disease-classification-main > api > main.py
11:16 LF UTF-8 4 spaces Python 3.12 (potato-disease-classification-main)

Home Workspaces Explore Search Postman Sign In Create Account

Working locally in Scratch Pad. **Switch to a Workspace**

Scratch Pad New Import

Collections APIs Environments Mock Servers Monitors History

You don't have any collections

Collections let you group related requests, making them easier to access and run.

Create Collection

Overview POST http://localhost:8000/ POST http://localhost:8000/ No Environment

http://localhost:8000/predict Save

POST http://localhost:8000/predict Send

Params Authorization Headers (10) Body Pre-request Script Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL

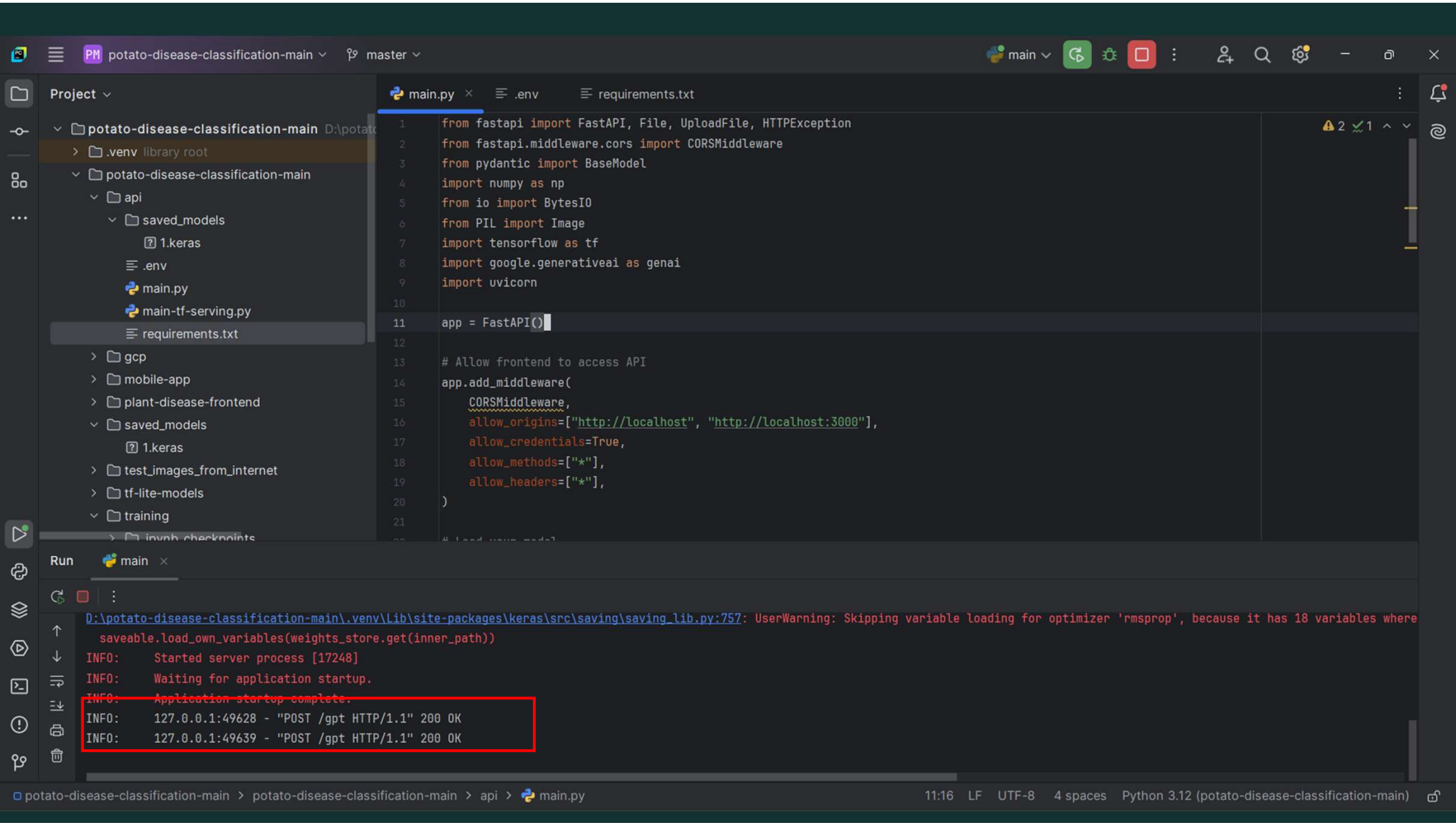
KEY	VALUE	DESCRIPTION	Bulk Edit
file	0ad3ba53-f01b-403b-a99d-5991eed85045___RS_Ea...		
Key	Value	Description	

Body Cookies Headers (4) Test Results Status: 200 OK Time: 14.38 s Size: 3.38 KB Save Response

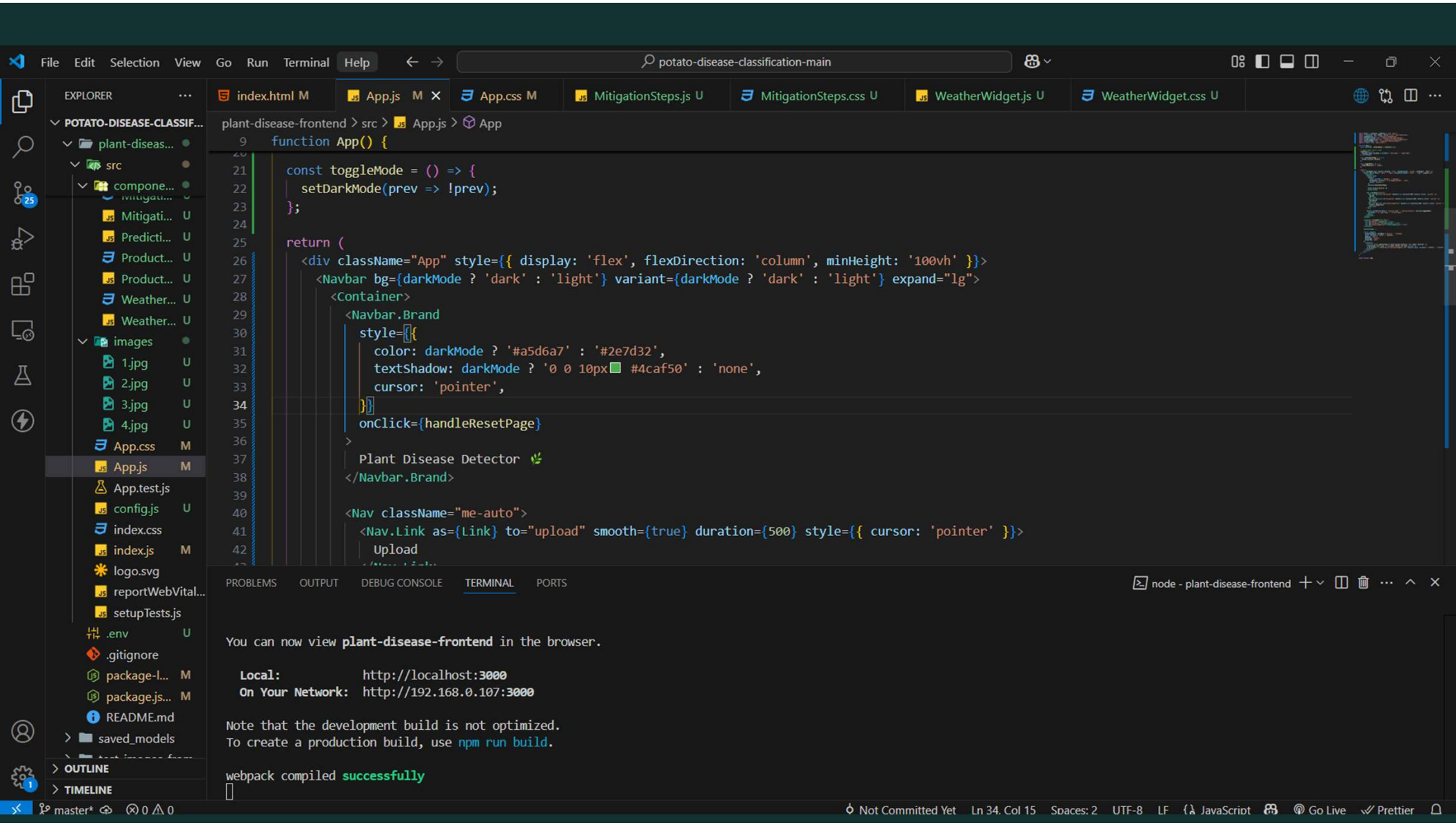
Pretty Raw Preview Visualize JSON

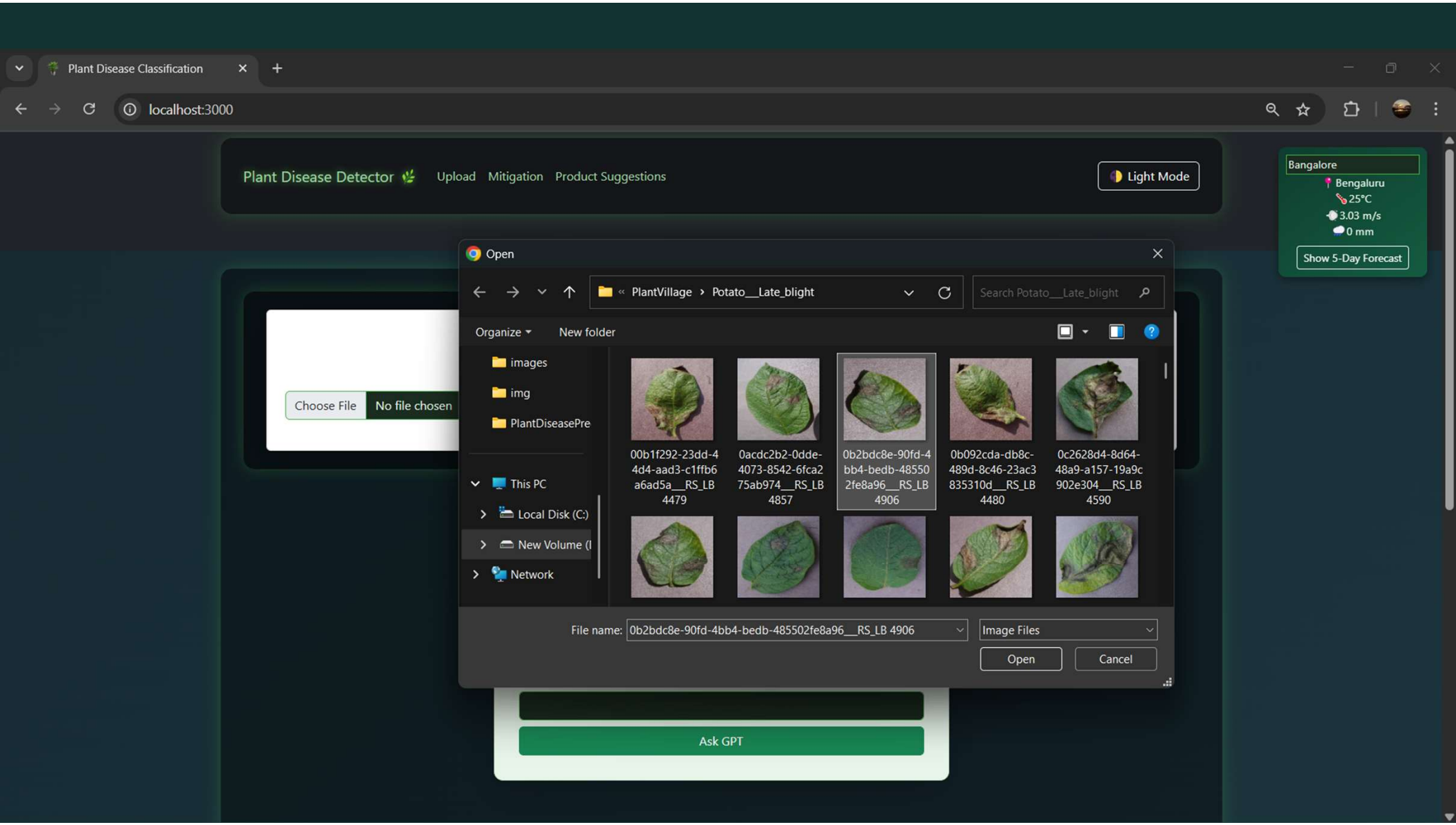

```
{
  "class": "Early Blight",
  "confidence": 0.9999991655349731,
  "advice": "## Early Blight Mitigation Steps in Tomatoes\n\nEarly blight (caused by *Alternaria solani*) is a common fungal disease
    affecting tomatoes. Here are the best mitigation steps, combining cultural practices and chemical controls:\n\n**Cultural Practices
    (Most Important & Preventative):**\n\n1. **Crop Rotation:** Rotate tomato crops every 2-3 years with non-solanaceous plants (not
    peppers, potatoes, eggplant).  This breaks the disease cycle.\n2. **Resistant Varieties:** Choose tomato varieties with some
    resistance to early blight.  Look for descriptions that mention resistance to *Alternaria*.\n3. **Proper Spacing:** Adequate spacing
    between plants allows for good air circulation, which helps leaves dry faster, reducing disease development.\n4. **Watering
    Practices:** Water at the base of the plants to avoid wetting the foliage. Overhead watering creates ideal conditions for fungal
    growth. Soaker hoses or drip irrigation are excellent choices.\n5. **Mulching:** Mulching around plants reduces soil splashing.
```


Find and Replace Console Runner Trash


potato-disease-classification-main
master
main
Project
potato-disease-classification-main D:\potato
.venv library root
potato-disease-classification-main
api
saved_models
1.keras
.env
main.py
main-tf-serving.py
requirements.txt
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allow_credentials=True,
allow_methods=["*"],
allow_headers=["*"],
)
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main
D:\potato-disease-classification-main\.venv\Lib\site-packages\keras\src\saving\saving_lib.py:757: UserWarning: Skipping variable loading for optimizer 'rmsprop', because it has 18 variables where
saveable.load_own_variables(weights_store.get(inner_path))
INFO: Started server process [17248]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:49628 - "POST /gpt HTTP/1.1" 200 OK
INFO: 127.0.0.1:49639 - "POST /gpt HTTP/1.1" 200 OK
potato-disease-classification-main > potato-disease-classification-main > api > main.py
11:16 LF UTF-8 4 spaces Python 3.12 (potato-disease-classification-main)

FRONT-END


File Edit Selection View Go Run Terminal Help
potato-disease-classification-main
EXPLORER
index.html M
App.js M
App.css M
MitigationSteps.js U
MitigationSteps.css U
WeatherWidget.js U
WeatherWidget.css U
plant-disease-frontend > src > App.js > App
function App() {
const toggleMode = () => {
setDarkMode(prev => !prev);
};
return (
<div className="App" style={{ display: 'flex', flexDirection: 'column', minHeight: '100vh' }}>
<Navbar bg={darkMode ? 'dark' : 'light'} variant={darkMode ? 'dark' : 'light'} expand="lg">
<Container>
<Navbar.Brand
style={{
color: darkMode ? '#a5d6a7' : '#2e7d32',
textShadow: darkMode ? '0 0 10px #4caf50' : 'none',
cursor: 'pointer',
}}
onClick={handleResetPage}
>
Plant Disease Detector
</Navbar.Brand>
<Nav className="me-auto">
<Nav.Link as={Link} to="upload" smooth={true} duration={500} style={{ cursor: 'pointer' }}>
Upload
POTATO-DISEASE-CLASSIF...
plant-diseas...
src
compone...
Mitigati... U
Predicti... U
Product... U
Product... U
Weather... U
Weather... U
images
1.jpg U
2.jpg U
3.jpg U
4.jpg U
App.css M
App.js M
App.test.js
config.js U
index.css
index.js M
logo.svg
reportWebVital...
setupTests.js
.env U
.gitignore
package-l... M
package.js... M
README.md
saved_models
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
node - plant-disease-frontend
You can now view plant-disease-frontend in the browser.
Local: http://localhost:3000
On Your Network: http://192.168.0.107:3000
Note that the development build is not optimized.
To create a production build, use npm run build.
webpack compiled successfully
master* 0 0
Not Committed Yet Ln 34, Col 15 Spaces: 2 UTF-8 LF JavaScript Go Live Prettier


Plant Disease Classification
localhost:3000
Plant Disease Detector
Upload
Mitigation
Product Suggestions
Light Mode
Bangalore
Bengaluru
25°C
3.03 m/s
0 mm
Show 5-Day Forecast
Choose File
No file chosen
Open
PlantVillage › Potato___Late_blight
Search Potato___Late_blight
Organize
New folder
images
img
PlantDiseasePre
This PC
Local Disk (C:)
New Volume (I
Network
00b1f292-23dd-44d4-aad3-c1ffb6a6ad5a___RS_LB 4479
0acdc2b2-0dde-4073-8542-6fca275ab974___RS_LB 4857
0b2bdc8e-90fd-4bb4-bedb-485502fe8a96___RS_LB 4906
0b092cda-db8c-489d-8c46-23ac3835310d___RS_LB 4480
0c2628d4-8d64-48a9-a157-19a9c902e304___RS_LB 4590
File name: 0b2bdc8e-90fd-4bb4-bedb-485502fe8a96___RS_LB 4906
Image Files
Open
Cancel
Ask GPT

Plant Disease Classification

localhost:3000

Plant Disease Prediction



Clear

Prediction: Late Blight

Confidence: 99.99%

Expert Advice

Mitigation Steps for Late Blight in Tomato Plants:

Late blight, caused by the water mold Phytophthora infestans, is a devastating disease. Prevention is key, as once established, it's incredibly difficult to control. Here are the best mitigation steps:

Cultural Practices:

- ***Resistant Varieties:*** *Choose tomato varieties with resistance to late blight (e.g., 'Defiant PhR', 'Mountain Magic'). This is your first and best line of defense.*
- ***Crop Rotation:*** *Rotate crops, avoiding planting tomatoes, potatoes, and other members of the nightshade family in the same location for at least 3 years.*
- ***Proper Spacing:*** *Ensure adequate spacing between plants for good air circulation, which helps reduce humidity and leaf wetness.*
- ***Watering Practices:*** *Water at the base of the plants, avoiding overhead watering that splashes spores onto leaves. Water early in the day so foliage can dry before evening.*

Bangalore

Bengaluru

25°C

3.03 m/s

0 mm

Show 5-Day Forecast

Plant Disease Classification

localhost:3000

Bangalore
Bengaluru
25°C
3.03 m/s
0 mm
Show 5-Day Forecast

Plant Doctor Bot

🧠 GPT Advice:

Hello! As a plant disease expert, my focus is on preventing problems rather than just curing them. Healthy plants are much more resistant to disease. Here are some tips for achieving good crops by minimizing disease risk:

1. Start with healthy soil:

* **Soil Testing:** Before planting anything, get your soil tested. This tells you the pH level and nutrient content. Amend the soil based on the results to create an optimal environment. The correct pH and balanced nutrients enhance plant vigor and disease resistance

Hello give me some tips for good crops

Ask GPT

🛒 Recommended Products



Neem Oil Bio-Pesticide
₹299
Buy Now



Copper Fungicide Spray
₹349
Buy Now



Organic Plant Protector
₹399
Buy Now



Fungal Disease Control Pack
₹449
Buy Now

Thank You!

We appreciate your time and interest in our project.

For questions or collaboration, please reach out via email or GitHub.

1 **Questions?**

We welcome your inquiries and feedback.

2 **Contact**

Email: nehakumari4911@gmail.com

3 **GitHub**

Explore code & updates at github.com/NehaKumari98